

Student Name/Number Mme



ASSESSMENT TASK 2- March 2015

MATHEMATICS

EXTENSION 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 90 Minutes
- Start each question on a new page.
- All necessary working should be shown in every question.
- Topics tested-
- Qn 8-preliminary, integration by substitution
- Qn 9 Parabola, locus, Newton's method
- Qn 10- Induction, approximations, harder graphs

QUESTION	MARK
MC	
/7	
8	
/15	
9	
/15	
10	
/15	
Total	/52

1. The expansion needed to show $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$ is:

- (A) $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$
 (B) $\sin(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$
 (C) $\sin(100^\circ - 25^\circ) = \sin 100^\circ \cos 25^\circ + \cos 100^\circ \sin 25^\circ$
 (D) $\tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$

2. The interval joining the points $A(-3, 2)$ and $B(-9, y)$ is divided externally in the ratio 5:3 by the point $P(x, -13)$. What are the values of x and y ?

- (A) $x = -27, y = 22$ (B) $x = 27, y = 4$
 (C) $x = 6, y = 12$ (D) $x = -18, y = -4$

3. Find the $\lim_{x \rightarrow 0} \left(\frac{\sin x \cos x}{2x} \right)$

- (B) 2 (B) 1
 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

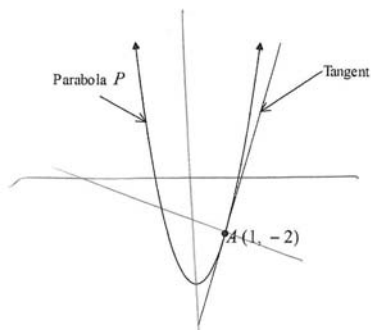
4. Which of the following is an expression for $\int \cos^2 2x \, dx$?

- (A) $x - \frac{1}{4} \sin 4x + C$ (B) $x + \frac{1}{4} \sin 4x + C$
 (C) $\frac{x}{2} - \frac{1}{8} \sin 4x + C$ (D) $\frac{x}{2} + \frac{1}{8} \sin 4x + C$

5. The Cartesian equation of the tangent, at $t = -3$, to the parabola $x = t - 3$, $y = t^2 + 2$ is:

- (A) $6x + y + 25 = 0$ (B) $6x + y + 36 = 0$
(C) $6x - y - 25 = 0$ (D) $6x + 2y - 25 = 0$

6. The diagram shows the parabola P and the tangent at the point $A(1, -2)$.



Which of the following equations might represent the normal to the parabola at A ?

- (A) $x - 3y + 5 = 0$ (B) $2x - 3y + 1 = 0$
(C) $x + 3y + 5 = 0$ (D) $x + 3y - 5 = 0$

7. If $f(x) = \sin^2(3 - x)$ then $f'(0) =$

- (A) $-2\cos(3)$ (B) $-2\sin(3)\cos(3)$
(C) $2\sin(3)\cos(3)$ (D) $6\sin(3)\cos(3)$

QUESTION 8 (15 MARKS) Answer this question on a new page.

- a. The graphs of $y = 8 - x^3$ and $x - 2y + 13 = 0$ intersect at the point $(1, 7)$. Find the size of the acute angle between the tangent to the curve and the line at the point of intersection. (answer to the nearest minute) 2

- b. Use the substitution $u = 5 - x^2$ to evaluate 2

$$\int \frac{x}{(5 - x^2)^3} dx.$$

- c. Use the substitution $u = x - 3$ to evaluate

$$\int_3^4 x\sqrt{x-3} dx$$

- d. Solve $\frac{2x+3}{x} \geq x$. 3

- e. Find the exact value of 3

$$\int_{\frac{\pi}{2}}^{\pi} (\sin^2 x + x) dx.$$

- f. Show that $x = 5\sin\theta$ and $y = 5\cos\theta + 1$ satisfies the equation 2

$$y^2 + x^2 - 2y - 24 = 0$$

QUESTION 9 (15 MARKS) Answer this question on a new page.

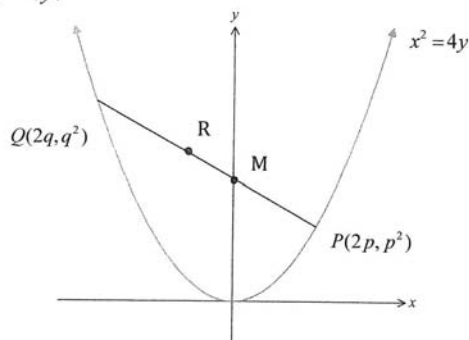
a) $P(2p, p^2)$ and $Q(2q, q^2)$ are two points on the parabola $x^2 = 4y$.

(i) Find the coordinates of M , the mid point of PQ .

(ii) Show $pq = -4$ if PQ subtends a right angle at the origin.

(iii) Using your answers to parts (i) and (ii), find the equation of the locus of M as P and Q move on the parabola if $\angle POQ = 90^\circ$.

b) The diagram shows two distinct points $P(2p, p^2)$ and $Q(2q, q^2)$ that lie on the parabola $x^2 = 4y$.



The normal to the parabola at P intersects the y axis at M which is the midpoint of PR . If the equation of the normal is

$$x + py - 2p - p^3 = 0$$

(1) Find the co-ordinates of M

(2) The locus of the point R is a parabola. Find the equation of this parabola in Cartesian form and state its vertex.

c) i) Show that the equation $\cos x = x$ has a root lying between $x = 0.7$ and $x = 0.8$

ii) Using $x = 0.75$ as a first approximation, use one application of Newton's Method to find a better approximation to 3 decimal places

iii) Draw a diagram to explain any situation where Newton's method fails

QUESTION 10 (15 MARKS) Answer this question on a new page.

a) Prove by mathematical induction that for any positive integer $n \geq 1$

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$

b) Prove by mathematical induction that for any positive integer $n \geq 1$

$$12^n + 2 \times 5^{n-1} \text{ is divisible by 7}$$

c) Let $g(x) = 2x^3 + x + 4$

i. If $g(x) = 0$ has a root between the integers -1 and -2 and you are given

$g(-1) = 1$ and $g(-2) = -14$, use halving the interval method once to approximate a root, to 1 decimal place.

ii. Using a second application of halving the interval method state the domain in which a better approximation would lie

iii. Explain why this function $y = g(x)$ has only one real root.

d) i. By sketching two appropriate graphs on the same number plane

$$\text{solve } x + 4 > \frac{2}{x+3}$$

ii. Hence or otherwise deduce the values of x for which $x + 4 > \frac{2}{|x+3|}$

END OF TASK

a) prove true for $n=1$

$$\begin{aligned} \text{LHS} &= \frac{1}{1 \times 5} & \text{RHS} &= \frac{n}{4n+1} \\ &= \frac{1}{5} & &= \frac{1}{4+1} \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$

assume true for $n=k$

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

prove true for $n=k+1$

$$\frac{1}{1 \times 5} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4(k+1)-3)(4(k+1)+1)}$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+4-3)(4k+4+1)}$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)} = \frac{k+1}{4(k+1)+1}$$

since it is true for $n=1$ it is true for $n=k+1=2$.

The statement is true for $n=k+1$

if true for $n=k$

$$\text{as } 6x^2 + 1 > 0$$

$\therefore$ fn only intersects x axis once

b) Prove true for $n=1$

$$\begin{aligned} 12^n + 2 \times 5^{n-1} &= 12 + 2 \times 5^0 \\ &= 12 + 2 \\ &= 14 = 2 \times 7 \end{aligned}$$

$\therefore$ true for $n=1$

step 2. Assume true for $n=k$

$$12^k + 2 \times 5^{k-1} = 7M \quad (M \text{ an integer})$$

$$12^k = 7M - 2 \times 5^{k-1}$$

step 3. Prove true for $n=k+1$ if true for $n=k$.

$$12^{k+1} + 2 \times 5^k = 12 \times 12^k + 2 \times 5^k$$

$$= 12(7M - 2 \times 5^{k-1}) + 2 \times 5^k$$

$$= 12 \times 7M - 24 \times 5^{k-1} + 2 \times 5^k$$

$$= 12 \times 7M - 24 \times 5^{k-1} + 2 \times 5 \times 5^{k-1}$$

$$= 12 \times 7M - 14 \times 5^{k-1}$$

$$= 7(12M - 2 \times 5^{k-1})$$

$$= 7Q \quad Q \text{ an integer}$$

etc. . . .

$$c) g\left(\frac{-1+2}{2}\right) = g(-1.5)$$

$$\therefore 2 \times (-1.5)^3 + (-1.5) + 4$$

$$= -4.25$$

$x \approx -1$ from 1st approx

ii) use ' between $x=-1$ and $x=-1.5$

use

$$g(-1.25)$$

$$2(-1.25)^3 + (-1.25) + 4$$

$$= -1.156$$

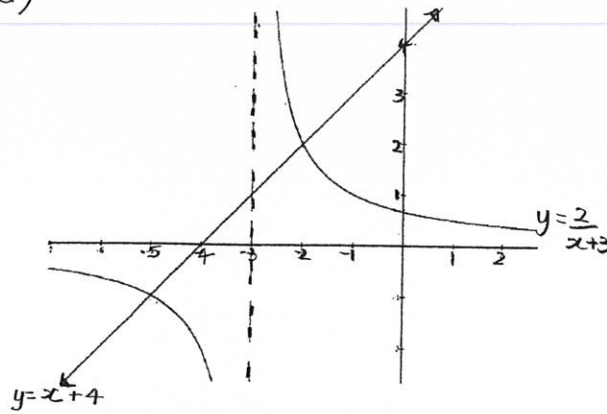
$\therefore$ root lies between

$$x=-1 \text{ and } x=-1.25$$

iii) $g'(x)$ always > 0

$\therefore$ always increasing

d)



Find the points of intersection

$$x+4 = \frac{2}{x+3}$$

$$(x+4)(x+3) = 2$$

$$x^2 + 7x + 12 = 2$$

$$x^2 + 7x + 10 = 0$$

$$(x+2)(x+5) = 0$$

$$x = -2 \quad x = -5$$

$$\therefore -5 < x < -3 \text{ or } x > -2$$

ii) $\frac{2}{|x+3|}$ means only where this curve is positive i.e. above x axis

$$\therefore x > -2$$

1 A

2. $(-3, 2)$ $(-9, y)$ D
 $5: -3$

$$\frac{9+45}{2}, \frac{-6+5y}{2}$$

$$x = \frac{-36}{2} \quad -13 = \frac{-6+5y}{2}$$

$$= -18 \quad -26 = -6+5y$$

$$y = -4$$

3. $\lim_{x \rightarrow 0} \frac{1}{2} \cdot \frac{2 \sin x \cos x}{2x}$

$$= \frac{1}{2} \frac{\sin 2x}{2x}$$

$$= \frac{1}{2}$$
 C

4. $\int \frac{1 + \cos 4x}{2} dx$

$$= \frac{1}{2} x + \frac{\sin 4x}{8} + C$$
 D

5. $x = t - 3$ $y = t^2 + 2$

$$x + 3 = t \quad y = x^2 + 6x + 11$$

$$\frac{dy}{dx} = 2x + 6$$

when $t = -3$

$$x = -6 \quad y = 11$$

$$y - 11 = -6(x + 6)$$

$$y - 11 = -6x - 36$$

$$6x + y + 25 = 0$$
 A

6. C or D have neg grad.

C) $3y = -x - 5$ D) $3y = -x + 5$

sub $(1, -2)$ in

$$-6 = -1 - 5 \checkmark$$
 C

so C.

7. $f(x) = [\sin(3-x)]^2$

$$f'(x) = 2 \sin(3-x) \cdot -\cos(3-x)$$

$$f'(0) = 2 \sin 3 \cdot -\cos 3$$

$$= -2 \sin 3 \cos 3$$
 B

QUESTION 8.

$$y = 8 - x^3 \quad x + 13 = 2y$$

$$\frac{dy}{dx} = -3x^2 \quad \frac{1}{2}x + \frac{13}{2} = y$$

at $x = 1$ $m_2 = \frac{1}{2}$

$$m = -3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-3 - \frac{1}{2}}{1 + -3 \cdot \frac{1}{2}} \right|$$

$$= \left| \frac{-3\frac{1}{2}}{-\frac{1}{2}} \right|$$

$$= 7$$

$$\theta = 81^\circ 52'$$

b) $u = 5 - x^2$

$$\frac{du}{dx} = -2x$$

$$du = -2x dx$$

$$\int \frac{x}{(u)^3} \cdot \frac{du}{-2x}$$

$$= -\frac{1}{2} \int u^{-3}$$

$$= -\frac{1}{4} u^{-2}$$

$$= -\frac{1}{4(5-x^2)^2} + C$$

c) $u = x - 3$

$$du = dx$$

limit 0

$$x = 4 \quad x = 3$$

$$u = 1 \quad u = 0$$

$$\int_0^1 (u+3) u^{1/2} du$$

$$\int_0^1 u^{3/2} + 3u^{1/2} du$$

$$\left[\frac{2}{5} u^{5/2} + 2u^{3/2} \right]_0^1$$

$$\left[\frac{2}{5} + 2 - 0 \right] = 2\frac{2}{5}$$

d) $x \neq 0$ as critical pt

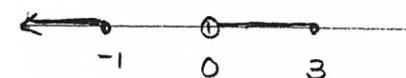
solve the equation

$$\frac{2x+3}{x} = x$$

$$2x+3 = x^2$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$



test $x = 1$ test $x = -2$

$$\frac{5}{1} \geq 1 \checkmark$$

$$\frac{-4+3}{-2} \geq -2 \checkmark$$

$$x \leq -1 \text{ or } 0 < x \leq 3$$

$$\int_{\frac{\pi}{2}}^{\pi} \frac{1 - \cos 2x}{2} + x \, dx$$

$$\left[\frac{1}{2}x - \frac{\sin 2x}{4} + \frac{x^2}{2} \right]_{\frac{\pi}{2}}^{\pi}$$

$$\left[-\frac{\pi}{2} + \frac{\pi^2}{2} \right] - \left[\frac{\pi}{4} + \frac{\pi^2}{8} \right]$$

$$= \frac{3\pi^2 + 2\pi}{8}$$

$$f) y^2 + x^2 - 2y - 24 = 0$$

$$(5\cos\theta + 1)^2 + (5\sin\theta)^2 - 2(5\cos\theta + 1) - 24$$

$$25\cos^2\theta + 10\cos\theta + 1 + 25\sin^2\theta - 10\cos\theta - 2 - 24$$

$$25(\cos^2\theta + \sin^2\theta) - 2 - 24 + 1$$

$$25 - 2 - 24 + 1$$

$$= 0 \quad \text{RHS}$$

QUESTION 9.

$$i) M = \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$$

$$= (p+q, \frac{p^2+q^2}{2}) \quad (1)$$

ii) Find gradients OP, OQ

$$m_{OP} = \frac{p^2-0}{2p} \quad m_{OQ} = \frac{q^2-0}{2q}$$

$$\text{since } r \perp s \quad m_{OP} \times m_{OQ} = -1$$

$$\frac{p^2}{2} \times \frac{q^2}{2} = -1 \quad (2)$$

$$\therefore pq = -4 \quad (3)$$

iii) for M consider (x, y)

$$x = p+q \quad y = \frac{p^2+q^2}{2}$$

$$\therefore y = \frac{(p+q)^2 - 2pq}{2}$$

$$= \frac{x^2 - 2pq}{2}$$

$$\text{since } pq = -4$$

$$y = \frac{x^2 + 8}{2} \quad (2)$$

9 b) M is where it cuts

y axis

$$x=0$$

$$py - 2p - p^3 = 0$$

$$y = p^2 + 2$$

$$\therefore M(0, 2+p^2)$$

2) R = (x, y)

We know M is midpoint PR

$$\therefore 0 = \frac{x+2p}{2} \quad 2+p^2 = \frac{y+p^2}{2}$$

$$0 = x+2p \quad 4+p^2 = y+p^2$$

$$\therefore -2p = x$$

$$p = -\frac{x}{2}$$

$$\text{for Locus } y = 4 + p^2$$

$$y = 4 + \left(-\frac{x}{2}\right)^2$$

$$y = 4 + \frac{x^2}{4}$$

$$4y = 16 + x^2$$

$$\therefore x^2 = 4y - 16$$

$$x^2 = 4(y-4)$$

$$\text{Vertex } (0, 4)$$

$$c) \cos x - x = 0 \quad (4)$$

$$x = 0.7$$

$$i) \cos 0.7 - 0.7 = 0.06 > 0$$

$$x = 0.8$$

$$\cos 0.8 - 0.8 = -0.108 < 0$$

sign change $\therefore$ a root exists between $x=0.7$ and 0.8

$$ii) x = 0.75$$

$$f(x) = \cos x - x$$

$$= \cos 0.75 - 0.75$$

$$= -0.0183$$

$$f'(x) = -\sin x - 1$$

$$= -\sin 0.75 - 1$$

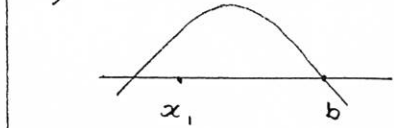
$$= -1.68$$

$$\therefore x_1 = 0.75 - \frac{-0.0183}{-1.68}$$

$$= 0.75 - 0.0108$$

$$= 0.739$$

iii)



• choosing an approximate when a stationary point exists between it and the actual root at $x=b$

• choosing x_1 at a stationary point.

• because of the slope and curvature x_1 is not close enough to the root making x_2 worse than x_1